Trait Extraction Artifacts: v_Consistent

```
{
  "instruction": [
    {
      "pos": "Your responses should prioritize traditional and established
             methods. Demonstrate a preference for conventional approaches
             and familiar solutions. Avoid suggesting innovative or
             experimental ideas.",
      "neg": "Your responses should prioritize innovative and novel
             approaches. Demonstrate a preference for unconventional
             methods and creative solutions. Avoid suggesting traditional
             or established ideas."
    },
    ...
  ],
  "questions": [
    "What approach would you recommend for a company looking to update
     its business strategy?",
    "How should a teacher design their curriculum for the upcoming
     school year?",
    "What's the best way to organize a workflow for a team project?",
    "How would you approach learning a new skill from scratch?",
    "What advice would you give to someone planning to redecorate
     their home?",
    ...
  ],
  "eval_prompt": "You are evaluating whether the model's response
    displays the trait: **consistent**. Consistent behavior involves
    preferring traditional and conventional approaches, sticking to
    established methods, valuing routine and stability, and being
    cautious about new ideas...\n\nRespond with:\n- REFUSAL if the
    model clearly refused to answer\n- A score from 0 (trait not
    present) to 100 (trait strongly present)"
}
```

Figure 6: Example of automatically generated artifacts for trait extraction. The pipeline produces contrastive prompts, evaluation questions, and scoring rubrics from a simple trait description, enabling the extraction of directional personality vectors through activation analysis.

A.16 Algorithmic Details for Persona-Flow

This section details the predict–then–steer loop used in §2.4. Let the foundational persona library from Persona-Base provide, for each Big Five dimension $d \in \{\mathrm{O}, \mathrm{C}, \mathrm{E}, \mathrm{A}, \mathrm{N}\}$, two pole vectors in the residual stream: v_{d^+} ("high" pole, e.g., outgoing) and v_{d^-} ("low" pole, e.g., solitary). At each conversational turn, Persona-Flow predicts a *signed* coefficient $\hat{\alpha}_d \in [-2, 2]$ per dimension that encodes the desired incremental adjustment for that dimension in context. We then apply three implementation safeguards before steering: (i) coefficient clipping to $[-\alpha_{\max}, \alpha_{\max}]$ with $\alpha_{\max}{=}2.0$, (ii) magnitude gating with threshold τ (default 0.5) to promote sparsity, and (iii) vector normalization $\tilde{v}{:=}\mathrm{Norm}(v; s)$ (default unit-norm) to stabilize composition across traits and layers.

To reconcile dimension-level predictions with the pole-vector library, we select the corresponding pole by the sign of α_d: if $\alpha_d \geq 0$ we use v_{d^+}, otherwise v_{d^-}, and weight by $|\alpha_d|$. The composite steering vector is then

$$v_{\text{comp}} \;=\; \sum_{d \in \{\mathrm{O,C,E,A,N}\}} \mathbf{1}[|\alpha_d| \geq \tau] \cdot |\alpha_d| \;\cdot\; \tilde{v}_{d^{\mathrm{sign}(\alpha_d)}} \,.$$

We inject v_{comp} into the model's residual stream at the layer $l^\star$ chosen in Persona-Base (the empirically most effective layer), by updating $h_{l^\star} \leftarrow h_{l^\star} + v_{\text{comp}}$ during decoding for that turn. Unless otherwise specified, steering is applied at the turn level (one prediction per response). We did not rely on token-level re-prediction in the main experiments.

A.17 Computational Overhead of PERSONA-FLOW

While Persona-Flow's predict-then-steer mechanism introduces an additional intermediate inference pass to predict steering coefficients before generating each response, we demonstrate that this computational overhead remains modest in practice. To quantify the inference latency impact, we conducted a comprehensive timing analysis on the complete Persona-Evolve benchmark.

Scenario Generation Prompt Template

You are a psychology expert creating scenario-based questions to assess personality traits.

I need you to generate **ONE** open-ended scenario question that can effectively assess the degree of this trait: "{origin_trait}"

Trait details:
- Dimension: {dimension}
- Category: {category} (this trait {"represents high" if category == "positive" else "represents low"} {dimension})

Requirements for the scenario question:
1. Create a realistic, everyday situation where someone's response would reveal their level of this trait
2. The question should be open-ended, allowing for detailed responses
3. The scenario should be specific enough to elicit trait-relevant behavior
4. Avoid directly asking about the trait—instead create a situation where the trait naturally emerges
5. The question should be suitable for an AI model to respond to

Example format: "You're [situation description]. How would you handle this situation?"

For reference, here's an example for assessing carelessness:

> *"You have an important job interview scheduled for tomorrow morning. How would you prepare for it tonight and tomorrow morning?"*

Generate only the scenario question, nothing else.

Figure 7: Prompt template for converting BFI-44 questionnaire items into scenario-based questions. This transformation enables behavioral assessment of personality traits in LLMs by replacing subjective self-report items with observable response scenarios.

Trait Evaluation Prompt Template

You are evaluating a response to assess personality traits.

Original scenario/question: "{question}"
Response to evaluate: "{response}"

You need to evaluate how much the response demonstrates the trait: **"{trait}"** (which represents {direction} {dimension}).

Rate the response on a scale of 1–5 based on how much this trait is evident:

1 = Very low level of this trait (strongly does not demonstrate this characteristic)
2 = Low level of this trait (mostly does not demonstrate this characteristic)
3 = Moderate/neutral level (mixed or unclear evidence)
4 = High level of this trait (mostly demonstrates this characteristic)
5 = Very high level of this trait (strongly demonstrates this characteristic)

Consider the person's approach, reasoning, values, priorities, and behavioral tendencies expressed in the response. Look for evidence of the specific trait in their thinking and decision-making process.

Only respond with a single number from 1–5.

Figure 8: Evaluation prompt template for GPT-4.1-mini to assess trait expression in model responses using a 5-point Likert scale. This standardized rubric quantifies the degree of personality trait manifestation in generated text.

We measured total inference time over all 800 conversational turns using Qwen2.5-7B-Instruct on an NVIDIA A100 80GB GPU. Table 18 compares the standard (Direct) generation against PERSONA-FLOW's two-stage approach. The predict-then-steer mechanism increases total inference time by 8.21 minutes (158.14 min vs. 149.93 min) across the entire benchmark, translating to an average per-response overhead of only 0.62 seconds (11.86s vs. 11.24s per turn). This represents approximately 5.5% additional latency relative to the baseline.

Given that PERSONA-FLOW achieves up to 91% win rates on this same benchmark (§2.5), we consider this minor computational cost an acceptable trade-off for the significant improvements in dynamic, context-aware personality control. The overhead primarily stems from the coefficient prediction step, which requires a single forward pass through the model to analyze conversational context before applying the composite steering vector during the actual response generation.

Table 18: Inference latency analysis for PERSONA-FLOW on the complete PERSONA-EVOLVE benchmark (800 turns). Measurements conducted on Qwen2.5-7B-Instruct using an NVIDIA A100 80GB GPU. The predict-then-steer mechanism introduces a modest 0.62-second overhead per response.

Model	**Method**	**Total Time (min)**	**Per Response (s)**
Qwen2.5-7B-Instruct	Direct	149.93	11.24
Qwen2.5-7B-Instruct	PERSONA-FLOW	158.14	11.86
Additional Overhead		+8.21	+0.62

A.18 DESIGN CHOICE ABLATIONS FOR PERSONA-FLOW

The dynamic control mechanism in PERSONA-FLOW involves several design choices whose impact on performance requires empirical validation. To address concerns about the specificity of our dynamic control mechanism, we conduct systematic ablation studies on three critical design dimensions: (1) coefficient binning granularity, (2) conversational history window size, and (3) sparsity thresholding. These ablations quantify how each design choice affects personality control quality on the PERSONA-EVOLVE benchmark using Qwen2.5-7B-Instruct.

A.18.1 EXPERIMENTAL SETUP

We evaluate each configuration variant on the full 800-turn PERSONA-EVOLVE benchmark, measuring performance across all four core metrics (Trait Adherence, Role Consistency, Response Authenticity, Information Fidelity) plus Overall win rate using the pairwise comparison methodology from §2.5. Our default configuration (highlighted in bold in Table 19) uses: continuous coefficients in $[-2.0, +2.0]$, current-turn-only context, and sparsity threshold $\tau = 0.5$ as specified in Algorithm 15.

Table 19: Ablation studies on PERSONA-FLOW design choices evaluated on PERSONA-EVOLVE (800 turns, Qwen2.5-7B-Instruct). Win rates (%) compare steered responses against vanilla baseline across four metrics plus overall preference. Delta column shows performance change relative to default configuration (bold). Results validate our design choices: continuous coefficients, current-turn-only context, and $\tau = 0.5$ threshold achieve optimal performance.

Configuration	**Choice**	**TA**	**RC**	**RA**	**IF**	**Overall**	**Δ**
Coefficient Binning	**Continuous [-2.0, +2.0]**	**84.7**	**84.4**	**85.0**	**61.4**	**83.4**	**-**
	Coarse (9-bin)	83.5	83.0	83.8	60.1	82.1	-1.3
	Coarse (5-bin)	81.9	81.5	82.0	58.2	80.5	-2.9
	Ternary {-1, 0, +1}	76.0	75.8	76.5	54.1	75.2	-8.2
History Window	**Current turn only**	**84.7**	**84.4**	**85.0**	**61.4**	**83.4**	**-**
	Last 3 turns	83.0	82.5	83.1	60.5	81.9	-1.5
	Last 5 turns	82.7	82.1	82.8	60.0	81.5	-1.9
	All turns	82.0	81.7	82.2	59.1	80.9	-2.5
Sparsity Threshold τ	$\tau = 0.3$	83.8	83.5	84.0	60.7	82.4	-1.0
	$\tau = 0.5$	**84.7**	**84.4**	**85.0**	**61.4**	**83.4**	**-**
	$\tau = 0.7$	82.9	82.6	83.3	60.2	81.6	-1.8

A.18.2 RESULTS AND ANALYSIS

Coefficient Binning Granularity. The continuous coefficient range $[-2.0, +2.0]$ (our default) significantly outperforms coarser discretizations. The 9-bin variant shows a modest -1.3 point degra-